Correlation

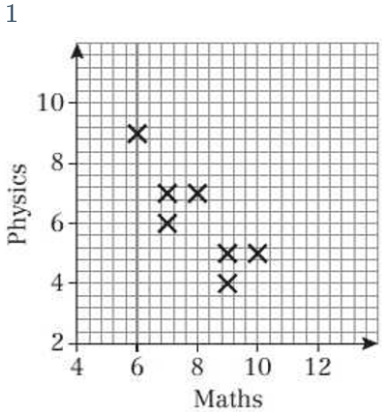
Starter

Solution 2

1) The table shows the scores out of 10 on a maths test and on a physics test for 7 students. Show this information on a scatter diagram.

Maths	6	7	7	8	9	9	10
Physics	9	7	6	7	5	4	5

1



2) A straight line has equation $y = 0.34 - 3.21x$.

Write down

a) The gradient of the line

2a -3.21

b) The y-intercept of the line

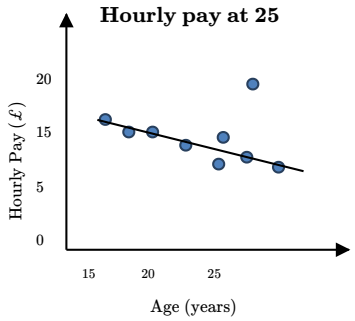
2b 0.34

This Chapter Overview

Previously we have only considered one variable at a time. When we introduce a second variable (e.g. height with age), **we might want to consider the relationship between them**.

This is a short chapter!

"Describe the type of correlation."



"The daily mean windspeed, w knots, and the daily maximum gust, g knots, were recorded. The equation of the regression line of g on w for these 15 days is $g = 7.23 + 1.82w$.

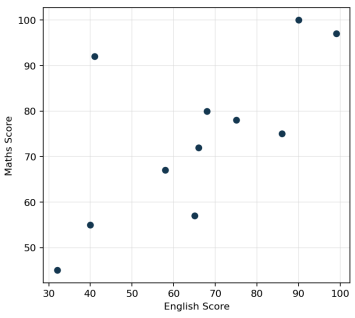
(a) Given an interpretation of the value of the gradient of this regression line.

(b) Justify the use of a linear regression line in this instance."

Changes since the old 'S1' syllabus:
This chapter has been scaled back significantly since the S1 'Correlation' and 'Regression' chapters. You no longer need to determine the equation of the line of best fit (the regression line), or calculate measures of correlation, but merely have to interpret an equation already given or the limitations of estimates made or comment on the suitability of a linear regression model.

Recap of correlation

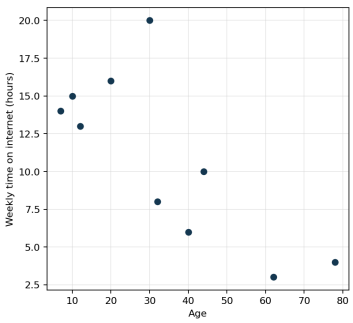
Correlation gives the **strength of the relationship** (and the type of relationship) between two variables. Data with two variables is known as **bivariate data**.



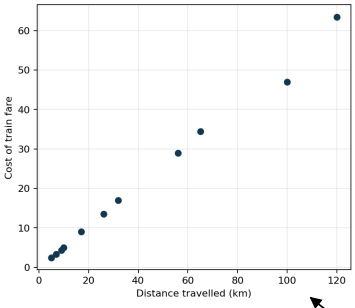
Type of correlation:

Weak positive correlation

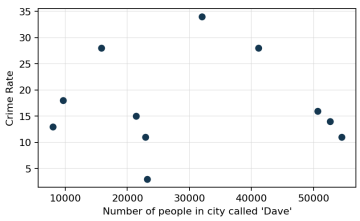
strength type



Weak negative correlation



Strong positive correlation



No correlation

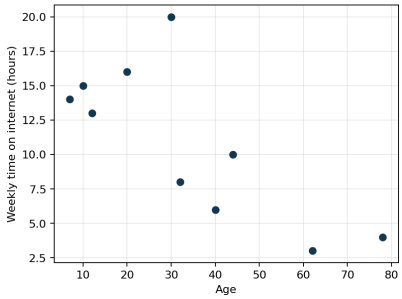
The vertical-axis variable usually **depends** on the horizontal-axis value. For this reason distance would be the **independent/explanatory variable** and cost the **dependent/response variable**.

Important correlation concepts

Solution 5

Important Point 1

To **interpret** the correlation between two variables is to give a worded description in the context of the problem.



a) State the correlation shown.
b) Describe/interpret the relationship between age and weekly time on the internet.

a) Negative correlation.

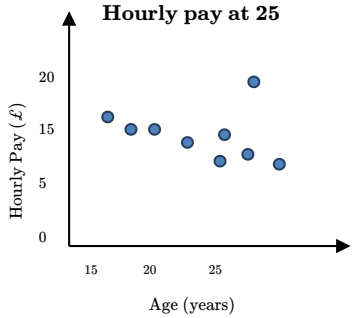
b) As age increases, the weekly time on the internet tends to decrease.

Important Point 2

[Textbook] Two variables have a **causal relationship** if a change in one variable directly causes a change in the other. Just because two variables show correlation it does not necessarily mean that they have a causal relationship.

Hideko was interested to see if there was a relationship between what people earn and the age which they left education or training. She says her data supports the conclusion that more education causes people to earn a lower hourly rate of pay. Give one reason why Hideko's conclusion might not be valid.

Respondents who left education later would have significantly less work experience than those who left education earlier. This could be the cause of the reduced income shown in her results.



Exercise 4A – Y1 Stats Pages 61-62

Solution 6

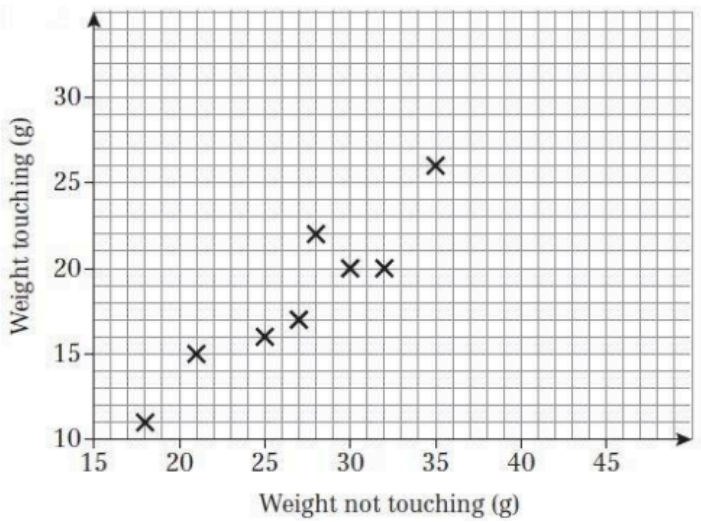
3. Eight students estimated the mass of a bag of sweets, in grams. First they estimated without touching the bag. They then picked it up and estimated the mass again.

Student	A	B	C	D	E	F	G	H
Without touching (g)	25	18	32	27	21	35	28	30
Holding the bag (g)	16	11	20	17	15	26	22	20

(a) Draw a scatter diagram to represent the data.

(b) Describe and interpret the correlation between the two variables.

a



b

Positive correlation. A student who guessed above (or below) the mean in one test tended to do the same in the other.

Exercise 4A – Y1 Stats Pages 61-62 — Solution 7

4. Donal investigated whether house values were related to internet speed, measured by the time taken to download a 100 megabyte file. The table shows his results.

Time taken (s)	5.2	5.5	5.8	6.0	6.8	8.3	9.3	13	13.6	16.0
House value (£1000s)	300	310	270	200	230	205	208	235	175	180

(a) Draw a scatter diagram to represent the data.

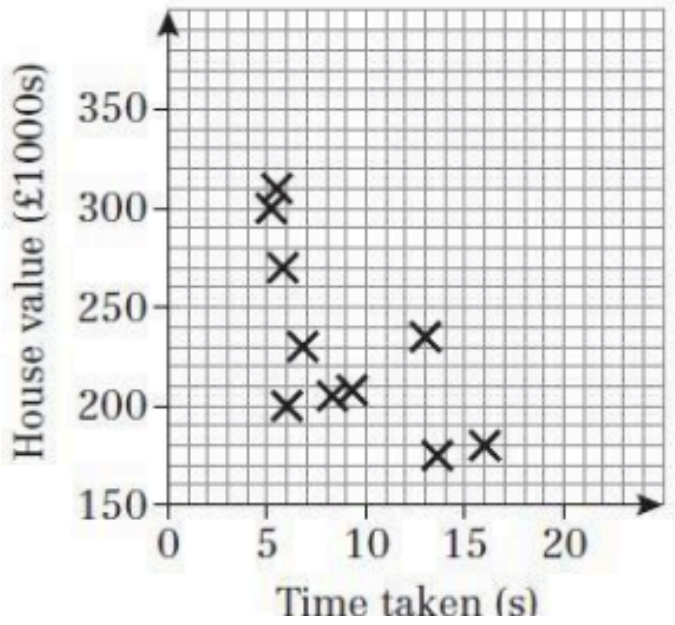
(b) Describe the type of correlation.

b Weak negative correlation.

(c) Donal says a slow connection reduces the value of a house. Give one reason why this conclusion may be invalid.

c
A third variable, such as distance from built-up areas, could influence both house value and internet speed.

a



Exercise 4A – Y1 Stats Pages 61-62 — Solution 8

5. Daily total rainfall r (mm) and sunshine s (hours) in Leuchars on 11 randomly selected days in August 1987, from the large data set.

r	0	6.8	0.9	4.8	0	21.7	1.7	4.9	0.1	2.2	0.1
s	8.4	4.9	10.2	4.5	3.3	3.9	5.4	1.8	9.7	1	4.6

The rainfall quartiles are $Q_1 = 0.1$, $Q_2 = 1.7$, $Q_3 = 4.85$. An outlier lies more than 1.5 interquartile ranges above the upper or below the lower quartile.

(a) Show that $r = 21.7$ is an outlier.

(b) Give a reason to (i) include and (ii) exclude this day's readings.

(c) Exclude this day and draw a scatter diagram for the remaining ten days.

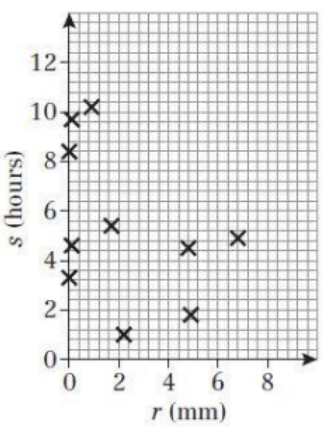
(d) Describe the correlation.

(e) Could the relationship between rainfall and sunshine be causal? Explain.

a $4.85 + 1.5(4.85 - 0.1) = 11.975 < 21.7$. It is an outlier.

b
(i) It is a genuine observation. (ii) It may be unrepresentative of typical rainfall.

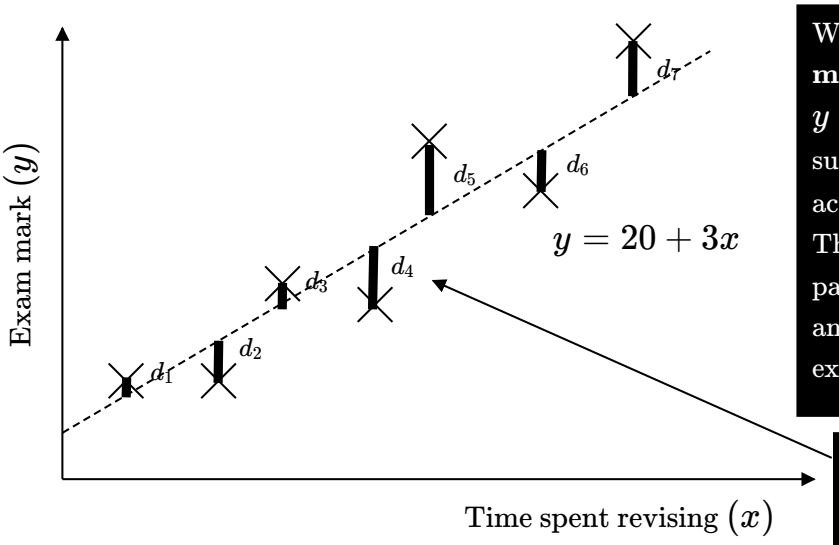
c



d Weak negative correlation.

e
There could be a causal relationship: rainy days tend to have more cloud and therefore less sunshine.

What is regression? Solution 9



What we've done here is come up with a **model** to explain the data, in this case, a line $y = a + bx$. We've then tried to set a and b such that the resulting y value matches the actual exam marks as closely as possible. The 'regression' bit is the act of setting the parameters of our model (here the gradient and y-intercept of the line of best fit) to best explain the data.

One type of line of best fit is the **least squares regression line**. This minimises the sum of the square of these 'errors', i.e.

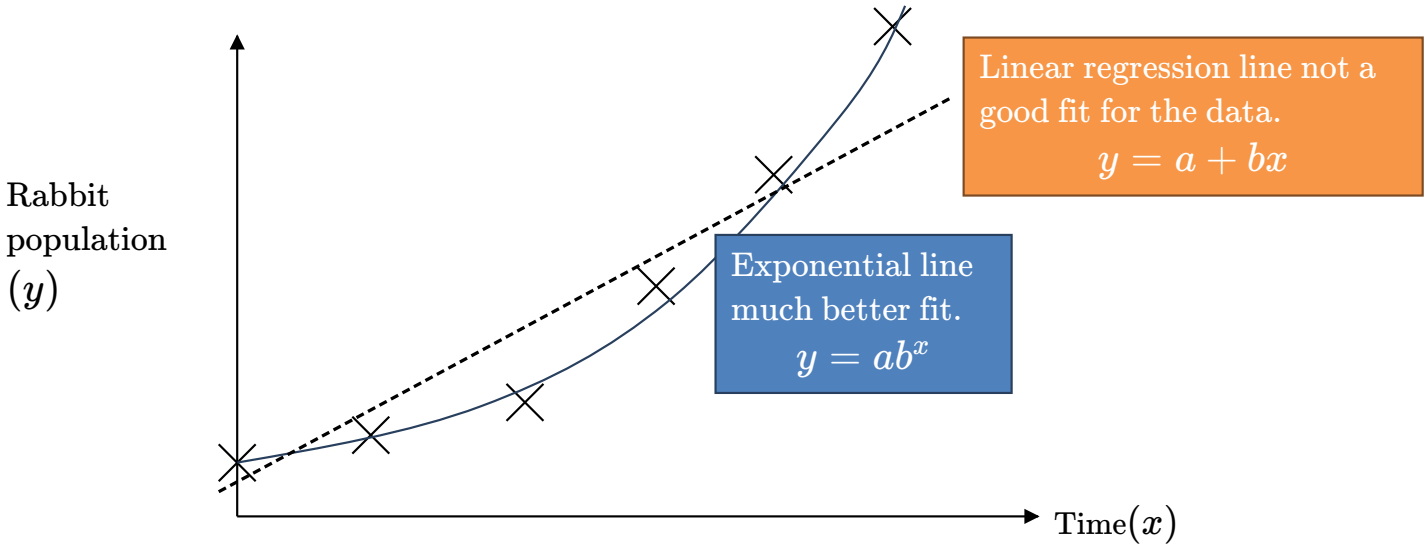
$$d_1^2 + d_2^2 + \ldots = \Sigma d_i^2$$

Part of the reason we square these errors is so that each distance is treated as a positive value.

Unlike in the old S1, you are no longer required to work out the equation of the least squares regression line yourself; you will be given the equation.

I record people's exam marks as well as the time they spent revising. I want to predict how well someone will do based on the time they spent revising. How would I do this?

What is regression? Solution 10

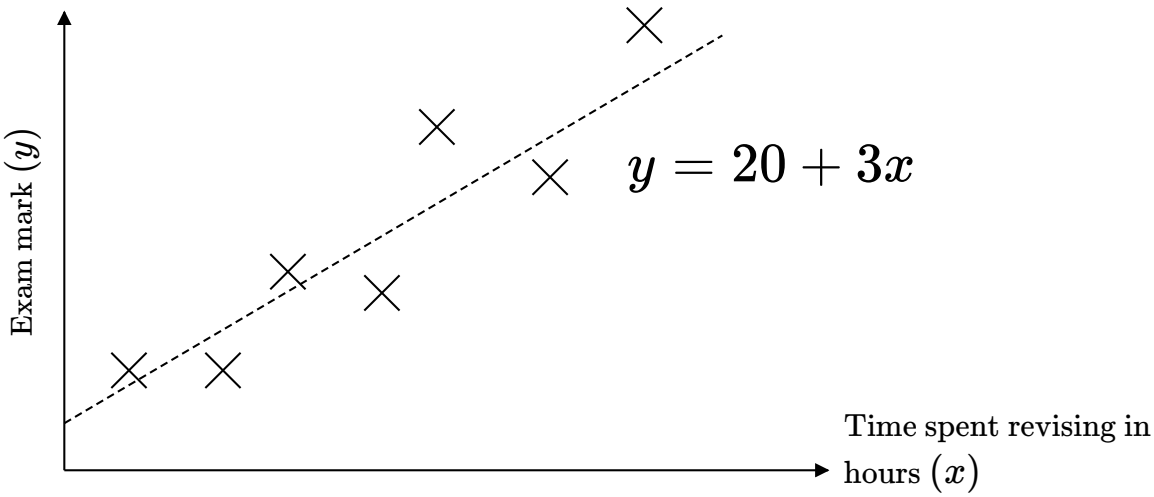


In this chapter we only cover **linear regression**, where our chosen model is a straight line.

But in general we could use any model that might best explain the data. Population tends to grow exponentially rather than linearly, so we might make our model $y = a \times b^x$ and then try to use regression to work out the best a and b to use. **You will do exponential regression in Chapter 14 of Pure Year 1.**

Interpreting a and b.

Solution 11



How do we interpret the gradient of 3?

For each extra hour spent revising, 3 marks are gained.
(i.e. the gradient tells you the change in y for each unit increase in x)

How do we interpret the y-intercept of 20?

20 marks would be obtained turning up to the exam with no revision.
(i.e. the value of y we get when $x = 0$)

Example

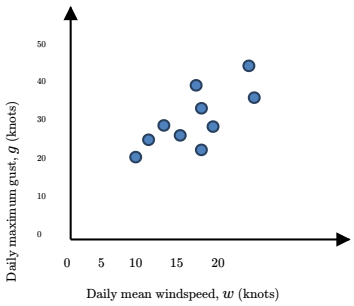
Solution 12

From the large data set, the daily mean windspeed, w knots, and the daily maximum gust, g knots, were recorded for the first 15 days in May in Camborne in 2015.

w	14	13	13	9	18	18	7	15	10	14	11	9	8	10	7
g	33	37	29	23	43	38	17	30	28	29	29	23	21	28	20

© Met Office

The data was plotted on a scatter diagram.



(a) Describe the correlation between daily mean windspeed and daily maximum gust.

The equation of the regression line of g on w for these 15 days is $g = 7.23 + 1.82w$

(b) Give an interpretation of the value of the gradient of this regression line.

(c) Justify the use of a linear regression line in this instance.

a There is a strong positive correlation between daily mean windspeed and daily maximum gust.

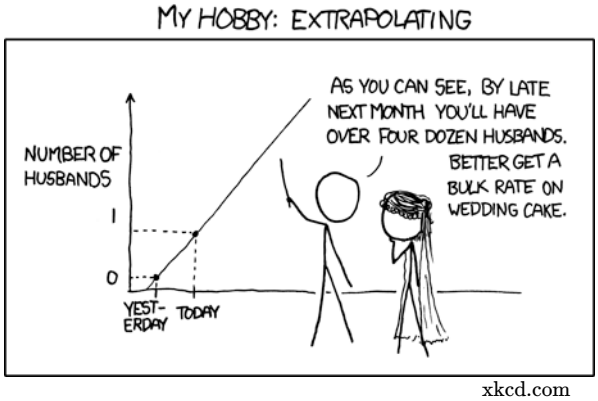
b If the daily mean windspeed increases by 10 knots the daily maximum gust increases by approximately 18 knots.

c The correlation suggests that there is a linear relationship between g and w so a linear regression line is a suitable model.

The stronger the (linear) correlation, the more suitable a linear regression line is.

Interpolating and Extrapolating

Solution 13



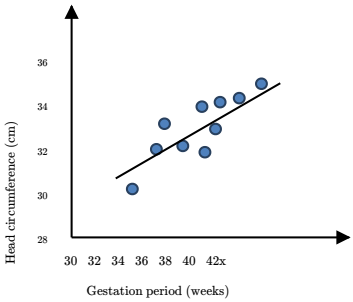
xkcd.com

You should only use the regression line to make predictions for values **of the dependent variable** that are **within the range of the given data**.

Estimating a value inside the data range is known as **interpolating**.
Estimating a value outside the data range is known as **extrapolating**
(as per the cartoon on the left!)

[Textbook] The head circumference, y cm, and gestation period, x weeks, for a random sample of eight newborn babies at a clinic are recorded.

The scatter graph shows the results.



The equation of the regression line of y on x is $y = 8.91 + 0.624x$. The regression equation is used to estimate the head circumference of a baby born at 39 weeks and a baby born at 30 weeks.

(a) Comment on the reliability of these estimates.

A nurse wants to estimate the gestation period for a baby born with a head circumference of 31.6cm.

(b) Explain why the regression equation given above is not suitable for this estimate.

a The prediction for 39 weeks is within the range of the data so is more likely to be correct.
The prediction for 30 weeks is outside the range of the data so is less likely to be accurate.

b The independent variable in this model is the gestation period, x. You should not use this model to predict a value of x for a given value of y.

Exercise 4B – Y1 Stats Pages 65-66

Solution 14

5. The large data set gives maximum relative humidity h (%) and mean visibility v (decametres) at Heathrow during the first two weeks of September 2015.

h	94	95	92	80	97	94	93	90	87	95	93	92	91	98
v	2600	2900	3900	4300	2800	2400	2700	3500	3000	2200	2200	3300	2800	2200

The regression line of v on h is $v = 12700 - 106h$.

(a) Interpret its gradient.

(b) Use your knowledge of the large data set to explain whether there is likely to be a causal relationship.

(c) Why would it be unreliable to predict (i) mean visibility at 100% humidity and (ii) humidity when visibility is 3000 decametres?

(d) State two ways to make better use of the large data set to model the relationship.

a For each percentage-point increase in humidity, predicted mean visibility falls by 106 decametres.

b
High humidity (above 95%) can cause mist and fog, reducing visibility. A causal link is plausible.

c
(i) 100% is outside the observed range, so the prediction is an extrapolation.
(ii) This line predicts visibility from humidity.
Predicting humidity requires the regression of h on v.

d
Randomly sample throughout September and analyse the whole month. Increase the sample size using several months between May and October.

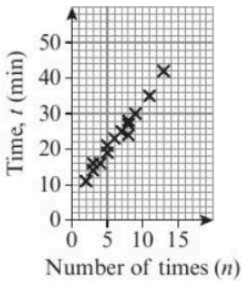
3. Packing times for a mail-order company employee were recorded for a random sample of orders.

Items (n)	2	3	3	4	5	5	6	7	8	8	8	9	11	13
Time (min)	11	14	16	16	19	21	23	25	24	27	28	30	35	42

(a) Describe the correlation between items packed and time taken. (1 mark)

The regression line is $t = 6.3 + 2.64n$.

(b) Interpret 2.64. (1 mark)



3a Strong positive correlation.

3b
Each additional item adds about 2.64 minutes to the predicted packing time.

6. James measured spring length s (cm) for attached masses m from 20 to 100 g. The regression line is $s = 44 + 0.2m$.

(a) Interpret 44 and 0.2. (2 marks)

(b) Why would it be unreasonable to use this equation to estimate (i) s when $m = 150$ or (ii) m when $s = 60$? (2 marks)

6a
With no mass attached, the predicted length is 44 cm.
Adding 1 g increases predicted length by 0.2 cm.

6b
(i) $m = 150$ is outside the data range, so this is extrapolation.

(ii) This line predicts s from m. To predict m from s, use the regression of m on s.